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Acute Kidney Injury in the Tropics

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Approximately 40% of the world population lives in the tropics, and this proportion is expected to rise to 60% by 2060. Tropical countries have high sociopolitical-economic heterogeneity, which influences health care. Facilities providing sophisticated care coexist with lack of basic public sector health infrastructure.[1-3]

Acute kidney injury (AKI) epidemiology in the tropics reflects these inequalities. In wealthy areas of the main cities and tertiary hospitals, AKI epidemiology is similar to that found in high-income countries in nontropical areas, and AKI principal causes are ischemia and drug nephrotoxicity. In contrast, in remote small cities, in vulnerable neighborhoods of large cities, and in rural zones, AKI is frequently community acquired, affecting previously healthy younger individuals. In those settings, infectious diarrhea, tropical infectious diseases, poisonous animals, natural medicines, and poor obstetric care are common causes of AKI.[1,2] Poverty, malnutrition, lack of quality sanitation, deficient public health systems, and uncontrolled urbanization are frequent and add significantly to the disease burden. Access to kidney replacement therapy (KRT) may be limited because of nonavailability and costs.[2-4] Epidemiologic data for community-acquired AKI in tropical countries showed a high frequency and high early mortality.[2-3]

There is regional variability in the cause of AKI in the tropics. Malaria, leptospirosis, diarrheal diseases, and venomous snakebite are common causes in South Asia. Malaria, diarrheal diseases, obstetric accidents, and indigenous herbal remedies are frequent in Africa. Malaria, leptospirosis, dengue fever, animal envenomation, and obstetric complications are important in Latin America.[1-4] Acute decreases in glomerular filtration rate (GFR) associated with dehydration from prolonged strenuous work in an unhealthy environment has been described in agricultural workers (see Chapter 66).[5] The long-term morbidity and mortality of tropical disease–associated AKI are nearly unknown.

SNAKEBITE

Snakebite is a neglected public health occupational hazard. Economically deprived populations living in rural communities are affected disproportionately. It has been suggested that snakebite envenomation causes 120,000 global deaths annually, with the highest burden in South and Southeast Asia, Latin America, and sub-Saharan Africa.[6] This is likely an underestimate because the data are based on hospital admissions. In 2019 the World Health Organization (WHO) released a strategy aiming to reduce snakebite-associated death and disability by 50% before 2030.[7]

Snake venoms are complex, with significant inter- and intraspecies variation. AKI is an important complication of snake venoms and may develop after bites by hemotoxic or myotoxic snakes belonging to the Viperidae and Elapidae families, such as Russell's viper, saw-scaled viper, puff adder, rattlesnake, tiger snake, green pit viper, *Bothrops, Lachesis,* South American rattlesnake *(Crotalus),* boomslang, gwardar, dugite, *Hypnale, Cryptophis,* and sea snakes. AKI is more frequent after Russell's viper bites (Fig. 71.1) in Asia and *Bothrops* and *Crotalus* bites in Latin America.[1,8-10] In a single center report from India, 21% of 866 snakebite patients over a 10-year period had developed AKI.[11] AKI prevalence is higher in children, probably because of the higher venom dose in relation to the body size.[1,10]

Clinical and Laboratory Features

Clinical manifestations depend on the nature and injected dose of venom. Local pain, swelling, blistering, ecchymosis and tissue necrosis at the bite site, and coagulation abnormalities leading to bleeding diathesis are frequent in Russell's viper and *Bothrops* bites (Figs. 71.2 and 71.3). Neurotoxicity (paralysis) and rhabdomyolysis (myalgia and cola-colored urine) may occur after sea snake and *Crotalus* bites.[1,8-11] Other uncommon manifestations include myocardial injury and hypopituitarism.[12,13]

AKI develops within a few hours (snake venom concentrates in kidney minutes after the bite) to as late as 96 hours after the accident.[1,8-11] The AKI is usually oliguric and catabolic. Oliguria generally lasts for 1 to 2 weeks, and its persistence suggests the likelihood of acute cortical necrosis, which can be confirmed by kidney biopsy or a contrast-enhanced computed tomography scan.[1,8-11] Although kidney function usually recovers, AKI-associated snakebite has been associated with chronic kidney disease (CKD) in both adults and children on long-term follow-up.[14] AKI after snakebite may aggravate kidney dysfunction in CKD of unknown ethology in agricultural workers.[15]

Laboratory investigation may disclose hemolysis (elevated free serum hemoglobin and lactate dehydrogenase and reduced haptoglobin), along with hypofibrinogenemia, reduced factors V, X, and XIIIA, protein C and antithrombin C, and elevated fibrin degradation products. Rhabdomyolysis may be present. Other findings include leukocytosis and elevated hematocrit as a result of hemoconcentration.[1,8-11] Laboratory features suggestive of thrombotic microangiopathy (TMA) were described in 19% cases,[16] related with severe and prolonged AKI and more likely to be associated with acute cortical necrosis. Elevations in urinary kidney structural biomarkers can predict AKI following Russell's viper bites.[17]

Pathology

Grossly, the kidneys may have petechial hemorrhages. On light microscopy, acute tubular cell injury ranging from mild changes to overt tubular necrosis, with hyaline or pigment casts, variable degree of interstitial edema and infiltration, and scattered hemorrhages are usually found. Mesangiolysis may occur (especially with *Crotalid* envenomation), and vessels may show fibrin thrombi. Electron microscopic findings include dense intracytoplasmic bodies representing degenerated organelles in the proximal tubules and electron-dense mesangial deposits. Less common findings are acute interstitial nephritis, necrotizing vasculitis,

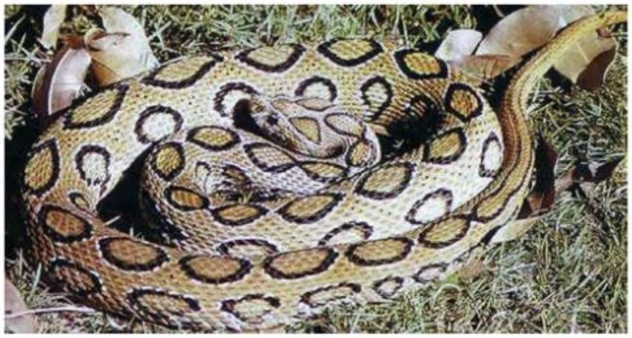

Fig. 71.1 Russell's Viper. This large snake is an important cause of venomous snakebite–induced acute kidney injury in Asia.

Fig. 71.2 Necrotic Finger Injury After *Bothrops* Snakebite. (Courtesy Carlos A.C. Mendes, São José do Rio Preto, Brazil.)

Fig. 71.3 Hemorrhagic Blister Developing a Few Hours After *Bothrops* Snakebite. (Courtesy Carlos A.C. Mendes, São José do Rio Preto, Brazil.)

and proliferative and crescentic immune-complex glomerulonephritis (GN). Acute cortical necrosis is seen in about 20% to 25% of cases after Russell's viper and *Echis carinatus* bites and has been described after *Bothrops* bites.[1,8-11]

Pathogenesis

Snake venom is a mixture of enzymes, toxins, peptides, carbohydrates, lipids, metals, biogenic amines, and nucleotides. Factors potentially involved in kidney injury include direct tubular toxicity, hemodynamic instability, hemolysis, rhabdomyolysis, systemic inflammation, glomerular deposition of microthrombi, oxidative stress, hyperuricemia, release of cytochrome c, and apoptosis of tubular epithelial cells.[1,8-11,18-20]

Management

Key steps include early administration of specific monovalent antivenom, appropriate volume replacement, maintenance of adequate urine output, correction of electrolyte abnormalities, administration of tetanus immunoglobulin, and treatment of infections.[1,8-11] Locally available polyvalent antivenom is effective against envenoming by multiple or unknown snakes.[21] Nonavailability of antivenom in rural hospitals and poor infrastructure that hampers transportation to health centers delay antivenom administration and may contribute to high mortality.[1,8-11,22] A recent systematic review highlighted the lack of high-quality evidence supporting the safety and efficacy of antivenom or other interventions in preventing or treating AKI.[23] Limited availability and the risk for side effects have encouraged the exploration of alternatives for antivenom therapies, but none has shown efficacy.[24] Therapeutic plasma exchange in cases with TMA was not effective.[16]

ARTHROPODS

Bees

Hundreds of simultaneous bee stings or multiple stings from wasps, yellow jackets, or hornets cause a multifaceted clinical picture comprising intravascular hemolysis; rhabdomyolysis; bleeding; cardiovascular, hepatic, and pulmonary injury; and severe AKI, with high mortality.[1,25,26] The potential mechanisms include direct venom nephrotoxicity, intrarenal vasoconstriction, hemoglobinuria, myoglobinuria, hypotension, TMA, and inflammatory and oxidative stress pathways activation.[27] Rarely, AKI can develop following a single sting, either due to anaphylactic reaction or secondary to TMA.[28] Kidney histology shows acute tubular necrosis.[1,27] The high morbidity and lethality of massive bee attacks has prompted the development of an antivenom, which was effective in protecting against rhabdomyolysis and hemolysis in mice.[29]

Caterpillars

Accidents with caterpillars of the genus *Lonomia* produce severe hemorrhagic disorders, characterized by both fibrinolytic and disseminated intravascular coagulation–like activity.[1] Severe and prolonged AKI, with some patients evolving to CKD, has been reported after *Lonomia obliqua* (Fig. 71.4) envenomation.[1,30] Early antivenom administration in rats prevented hemorrhagic manifestations and kidney dysfunction.[31] The AKI mechanisms likely involve glomerular deposition of fibrin, intravascular hemolysis, direct venom nephrotoxicity, activation of the endothelial cell inflammatory pathway, increased kidney expression of proteins involved in cell stress, heme-induced oxidative stress, coagulation, complement system activation, and kinin release.[32,33] Kidney histologic findings include acute tubular necrosis, ischemia, and tubular atrophy.[30,32]

Fig. 71.4 *Lonomia obliqua* **Caterpillars.** Each hair works as a miniature hypodermic needle to inject the hemolymph, which contains powerful venom that is able to induce severe coagulation system changes. (Courtesy Elvino J.G. Barros, Porto Alegre, Brazil.)

Fig. 71.5 *Loxosceles* **spp. (Brown Recluse Spider).** (Courtesy Katia C. Barbaro, São Paulo, Brazil.)

Loxosceles (Brown Recluse Spider)

Loxosceles genus spiders (Fig. 71.5) cause local necrosis at the bite site (Fig. 71.6), intravascular hemolysis, rhabdomyolysis, coagulation abnormalities, and AKI.[1,34] Even patients with mild cutaneous lesions may develop severe hemolysis and AKI, which is the main cause of death after these accidents. AKI pathogenesis has been related to massive intravascular hemolysis, kidney vasoconstriction, direct nephrotoxicity, and rhabdomyolysis.[1,34,35] Kidney histologic examination shows pigment-induced acute tubular necrosis.[34] Antivenom is likely the most effective therapy against AKI, but its use is frequently delayed because the bite goes unnoticed.

Scorpions

Scorpion venom causes autonomic overactivity and massive discharge of vasoactive substances and inflammatory cytokines and concentrates rapidly in kidney tissue.[36] Scorpion venom–associated AKI is rare.[37,38] AKI mechanisms are probably kidney vasoconstriction, hemodynamic instability, rhabdomyolysis, systemic inflammation, hemoglobinuria, mitochondrial dysfunction, and direct venom nephrotoxicity.[36-39] Kidney structural changes include acute tubular necrosis, glomerular changes, interstitial infiltrates, TMA, and acute cortical necrosis.[36,37,38,40]

NATURAL MEDICINE

Herbs and indigenous remedies are widely used in poor societies living in the tropics and are an important cause of AKI, especially in sub-Saharan Africa. This subject is discussed in Chapter 79.

MALARIA

Five species of *Plasmodium* parasites (*falciparum, vivax, malariae, ovale,* and *knowlesi*) cause malaria. Globally, malaria afflicts over 200 million people annually, causing nearly 500,000 deaths, mostly in children with *P. falciparum.* The contribution of malaria as the cause of AKI among different geographic areas ranges from 2% to 39%. AKI frequency in *P. falciparum* malaria reaches 60%, and in *P. vivax* malaria varies from 10% to 19%.[41] *P. malariae, P. knowlesi,* and *P. ovale*–induced AKI are less common. Malaria-associated AKI frequency using Kidney Disease: Improving Global Outcomes criteria is higher than using the WHO criteria.[42] In this chapter we focus on *P. falciparum* malaria–associated AKI.

Pathophysiology

P. vivax, P. ovale, and *P. knowlesi* infect young erythrocytes, and *P. malariae* infects aging cells. *P. falciparum* infects erythrocytes of all ages, producing a higher number of merozoites (Fig. 71.7). Heavy parasitemia is commonly observed in *P. falciparum* malaria, creating adverse effects on the microcirculation. Parasitized erythrocytes play a major role in the pathophysiologic process of the disease through decreased erythrocyte deformability and sequestration, knobs and rosette formation, cytoadherence, and changes in membrane transport and permeability (Fig. 71.8). The presence of knobs on parasitized erythrocyte membranes and cytoadherence between parasitized erythrocytes and vascular endothelial cells are characteristic of *P. falciparum* malaria. Cytoadherence of mature *P. vivax* parasitized erythrocytes to vascular endothelial cells involves intercellular adhesion molecule-1, CD36, endothelial protein C receptor, platelet endothelial cell adhesion molecule, and chondroitin sulfate A receptors.[43] *P. vivax*–caused cytoadherence is lower than that of *P. falciparum*–infected erythrocytes. Several proinflammatory cytokines (tumor necrosis factor; interleukins 1, 6, and 8; interferon-γ) and vasoactive mediators are released. The renin-angiotensin-aldosterone system is also stimulated. Hemodynamic changes in malaria are similar to those of bacterial sepsis, including decreased systemic vascular resistance, hypervolemia, increased cardiac output, and increased kidney vascular resistance. Because of increased vascular permeability, the initial hypervolemia is followed by hypovolemia and decreased cardiac output, with decreased kidney blood flow and GFR. Increased blood viscosity, intravascular coagulation, hemolysis, rhabdomyolysis, jaundice, fever, lactic acidosis, complement activation, and reactive oxygen species further compromise kidney blood flow. Activation of poly (adenosine diphosphate [ADP]-ribose) polymerase by peroxynitrite and free radicals decreases oxygen utilization.

AKI in malaria is ischemic and hypoxic in origin and usually occurs with heavy parasitemia, intravascular coagulation, intravascular hemolysis, or rhabdomyolysis.[41] Immune responses in malaria involve Th1 and Th2 activation. Immune complex GN can occur (see Chapter 57). Tubular changes vary from cloudy swelling to tubular degeneration (Fig. 71.9). Bile and hemoglobin casts and Tamm-Horsfall protein are present in the tubular lumen. Interstitial mononuclear infiltration and edema can occur. TMA and acute cortical necrosis have been reported.[44,45] Malarial antigens are occasionally seen along the glomerular endothelium and medullary capillaries. Adhesion molecules and proinflammatory cytokines are overexpressed in the vascular endothelium and proximal tubules.[41]

Fig. 71.6 **Local Necrotic Injury in the Left Leg of a Female Patient After *Loxosceles* Bite.** (A) At 4 days after the bite. (B) At 60 days after the bite. (C) At 3 months after the bite. (Courtesy Carlos A.C. Mendes, São José do Rio Preto, Brazil.)

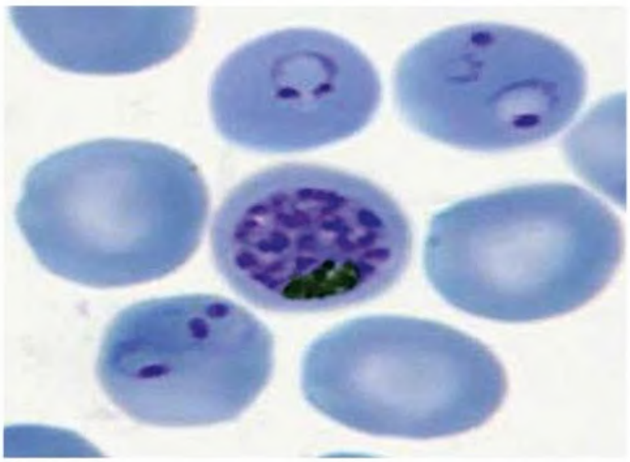

Fig. 71.7 Ring form and merozoites of *Plasmodium falciparum* in infected erythrocytes.

Clinical Manifestations

Malarial AKI affects predominantly nonimmune adults, mostly infected by *P. falciparum*. Reports link *P. vivax* infection with acute respiratory distress syndrome (ARDS), severe AKI, and TMA.[44] Mixed *Plasmodium* infections and comorbidities related to sepsis are likely important contributing factors to AKI. Symptoms include fever, chills, headache, prostration, and occasionally jaundice. The urinalysis usually shows few erythrocytes, leukocytes, and granular casts and proteinuria, often less than 1 g/24 hr. Hemoglobinuria happens in the patient with intravascular hemolysis, frequently associated with glucose-6 phosphate dehydrogenase deficiency. Rhabdomyolysis with myoglobinuria can be observed. Fluid and electrolyte changes are common in malaria.[41,46] Hyponatremia, usually asymptomatic, is observed in 67% of patients, is related to the severity of malaria, and resolves within a few days after antimalarial treatment. The causes of hyponatremia are multiple, including increased antidiuretic hormone with water retention, intracellular shift of sodium as a result of decreased Na^+,K^+-ATPase activity, and sodium depletion. Hypernatremia is uncommon, indicates hypothalamic lesions with diabetes insipidus, and is associated with unfavorable prognosis. Hypokalemia occurs in 20% to 40%

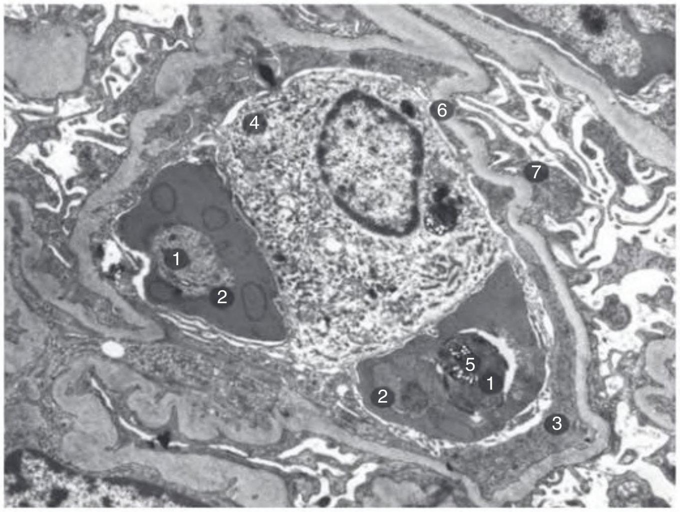

Fig. 71.8 Transmission electron micrograph from the kidney in *Plasmodium falciparum* malaria showing sequestration of two parasitized ed blood cells *(2)* within a glomerular capillary. Malarial pigment *(5)* is seen in a phagocyte *(4)*. *(1) P. falciparum.* *(3)* Glomerular endothelial cell. *(6)* Glomerular basement membrane. *(7)* Podocyte. (Courtesy Dr. Emsri Pongponratn, Faculty of Tropical Medicine, Mahidol University.)

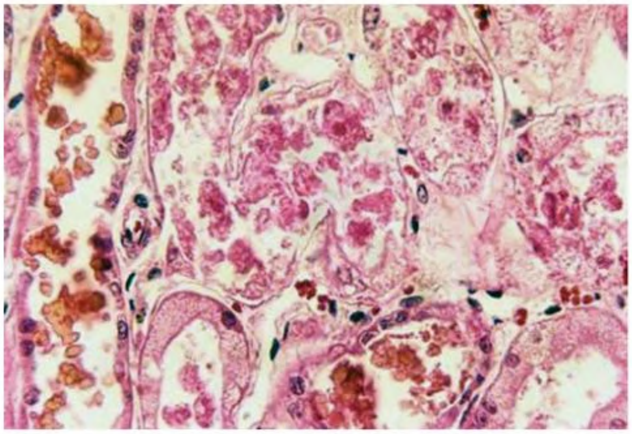

Fig. 71.9 Acute tubular necrosis in *Plasmodium falciparum* malaria.

of patients with uncomplicated cases. Hyperkalemia is observed in patients with intravascular hemolysis, rhabdomyolysis, and/or AKI. Hypocalcemia with prolonged QTc interval occurs in 45% of severe malaria cases and resolves with infection control.[46] Decreased activities of Na^+,K^+-ATPase, Ca^{2+}-ATPase, and parathyroid hormone are considered the main causes for hypocalcemia. Hypophosphatemia is observed in 6% to 30% and hypomagnesemia in 30% of patients.[46] Downregulation of sodium channels on the apical membrane of alveolar epithelial cells and increased vascular permeability account for the development of pulmonary edema and ARDS.[47]

Malarial AKI is characterized by a rapid increase in serum creatinine and is often associated with cholestatic jaundice. Risk factors for AKI include high parasitemia, jaundice, high liver enzymes, thrombocytopenia, decreased diastolic blood pressure, retinopathy, and ARDS. Hemolytic uremic syndrome has been described. Severe acidosis-associated lactic acidemia, hypoglycemia, and central nervous system symptoms are observed in severe malaria. Malarial AKI with multiple organ involvement and papilledema carries an unfavorable prognosis. AKI duration ranges from one to several weeks and is oliguric in 60% of the patients. Quinine and artesunate are the antimalarial agents most commonly used. The first choice in malarial AKI is artesunate, because it has high efficacy and does not need adjustment for kidney function. Early and frequent KRT (hemodialysis or peritoneal dialysis) is lifesaving.[41] Continuous venovenous hemofiltration yields good results in patients with multiorgan involvement, especially those with pulmonary edema or ARDS.[41] Exchange blood transfusion and erythrocytapheresis are adjunctive for the patient with heavy parasitemia.[48] The mortality rate of malarial AKI ranges from 10% to 50%. Kidney function may not recover completely in severe AKI, evolving to CKD in 7% of malaria-induced AKI.

LEPTOSPIROSIS

Leptospirosis, the world's most common zoonosis, is caused by *Leptospira* genus spirochetes.

The WHO included leptospirosis as a reemerging infectious disease in both higher and lower income areas, and cases have been increasingly reported in wealthy countries. Wild and domestic mammals (rodents, dogs, pigs, cattle, horses) are the disease vectors. The infection is transmitted accidently to humans through broken skin or mucosal membranes exposed to water or soil contaminated with organisms shed in the urine from the natural vectors.[1] Leptospirosis is a job-related threat for rodent exterminators, slaughterhouse workers, farmers, pet traders, veterinarians, garbage collectors, and sewer workers. Human leptospirosis is endemic in many tropical countries and usually reaches epidemic levels after heavy rainfall and flooding, or natural disasters, such as hurricanes or earthquakes. A meta-analysis revealed that flooding, lacerated wounds, and history of contact with livestock were independent risk factors for human leptospirosis.[49] There is an annual global estimate of 1 million cases, with 58,900 fatalities, with about half among young adult males.[50] A high seroprevalence of anti-*Leptospira* antibodies has been found in the general asymptomatic population in tropical countries.[1]

Leptospira interrogans, the only parasitic species, is mobile, aerobic, and unstained by the Gram method. Its endotoxins primarily affect tubulointerstitial cells. The bacterial outer membrane contains lipopolysaccharide, cytotoxic glycolipoprotein (GLP), and lipoproteins (LipL), especially LipL 32, which is immunogenic. Because *Leptospira* have special tropism for the kidneys, the effect of GLPs on tubular Na^+,K^+-ATPase activity is potentially involved in the AKI cellular pathophysiology, in the urinary concentrating defects, and in the paradoxical hypokalemia frequently seen in these patients.[1] Glycocalyx and endothelial injury caused by spirochete membrane proteins also may contribute to AKI (Fig. 71.10).

Kidney involvement is almost universal in leptospirosis but becomes relevant in Weil disease, the most severe form of leptospirosis, which is characterized by multiorgan involvement, with diffuse alveolar hemorrhage, pulmonary edema, ARDS, or a combination of these features, accompanied by AKI and a high mortality rate. Leptospirosis-associated AKI incidence varies from less than 10% to more than 80%, and its severity is associated with increasing mortality. Oliguria, jaundice, arrhythmias, thrombocytopenia, elevated baseline serum creatinine, and urinary neutrophil gelatinase–associated lipocalin levels have been associated with AKI development.[51] AKI is typically nonoliguric and associated with hypokalemia.[1] Tubular changes characterized by high urinary fractional excretion of sodium and potassium precede reduced GFR, which could explain the high frequency of hypokalemia. Long-term follow-up of leptospirosis-associated AKI found late development of CKD and presence of tubular function abnormalities.[52]

Early diagnosis and treatment are pivotal to successful leptospirosis treatment. The symptoms of leptospirosis (fever, myalgia, and headache) are nonspecific, jeopardizing its diagnosis.[53] The THAI

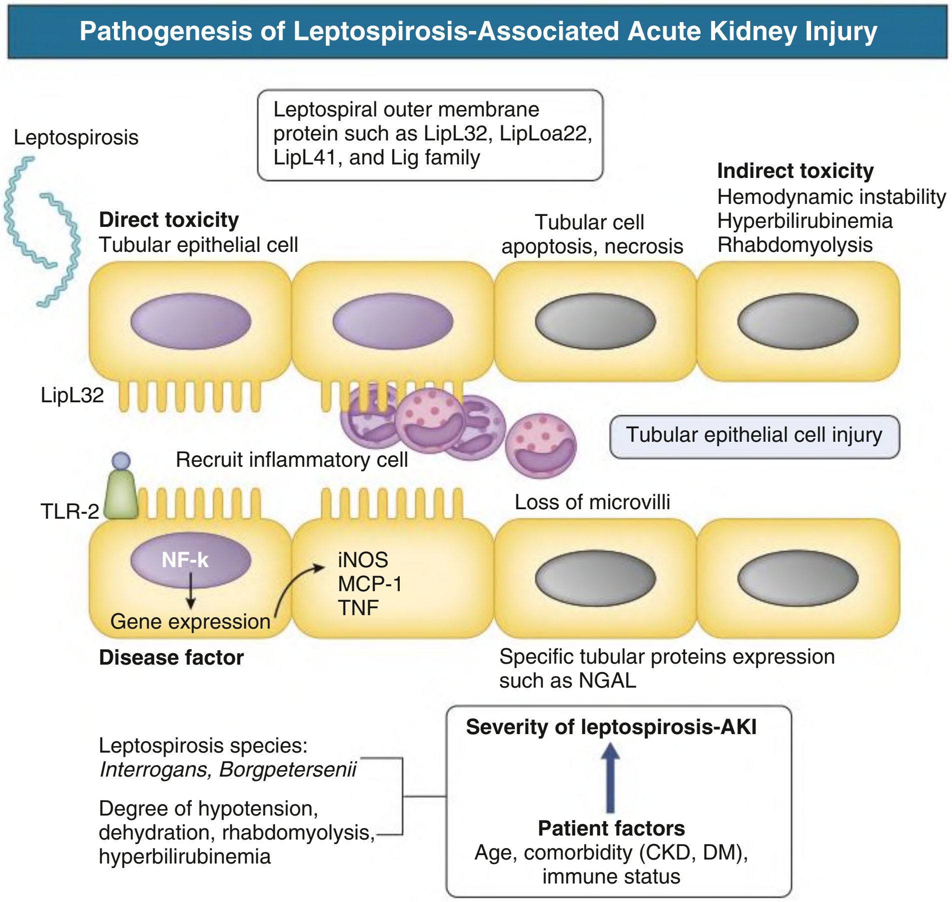

Fig. 71.10 Pathogenesis of leptospirosis associated acute kidney injury (AKI). *CKD,* Chronic kidney disease; *DM,* diabetes mellitus; *NGAL,* neutrophil gelatinase-associated lipocalin.

LEPTO score was proposed as a tool to help distinguish leptospirosis from other diseases in the inpatient setting. It includes hypotension, jaundice, muscle pain, AKI, low hemoglobin, hypokalemia with hyponatremia, and neutrophilia. The score showed a high discriminatory power with area under the curve (AUC) of 0.82.[54] A subsequent report in an outpatient department setting showed that the simplified score (Lepto OPD score) comprising history of exposure to wet ground at workplace, contact with water reservoir used by animal, positive urine dipstick test for protein and blood, and neutrophil count higher than 80% had an AUC of 0.72.[55] Microtranscriptome profiles of patients with severe leptospirosis versus nonsevere were different, suggesting novel biomarkers for severe leptospirosis.[56]

The leptospirosis diagnostic tests recommended by WHO include microscopic agglutination test, direct culture, and polymerase chain reaction technique. These techniques require advanced facilities, which are not universally available. Rapid diagnostic test kits, which mainly test serum immunoglobulin M antibodies, might be an alternative for this situation. However, different test kits have diversity of accuracy in different populations, and so their performance must be evaluated in specific populations before its use.[57]

Antibiotics of the penicillin group are considered the basis of treatment, but their efficacy remains unproven.[58] A study exploring the role of corticosteroid and desmopressin in severe leptospirosis patients with pulmonary involvement did not demonstrate any benefit on survival.[59] Current treatment recommendations include a high KRT dose, conservative fluid intake, and approaches to minimize lung injury, including extracorporeal membrane oxygenation support when indicated.[60]

HEMORRHAGIC FEVERS

Viral hemorrhagic fevers (VHFs) are caused by RNA viruses of four different families (Flaviridiae, Arenaviridae, Bunyaviridae, Filoviridae). Dengue, Rift Valley fever, yellow fever, and Crimean-Congo virus can be acquired by bites from infected arthropods, by inhalation of infected rodent excreta particles (Lassa, Junin, Machupo, Hantaan virus), or through contact with contaminated material (Ebola virus). The clinical picture includes fever, malaise, increased vascular permeability, and coagulation abnormalities that may lead to bleeding. Dengue and yellow fever are the most prevalent forms of VHF in the tropics.[1,61,62]

Dengue Fever

Dengue, which is currently the most important urban arboviral disease, is an acute febrile disease caused by an arbovirus transmitted primarily by mosquitoes, mainly the female *Aedes aegypti* mosquito. The number of yearly infections is estimated at 390 million.[62,63]

Several factors account for the rising incidence and widespread occurrence of dengue, such as climate changes (global warming, intensity and duration of rain season, hurricanes), ecosystem modifications, demographic increases, uncontrolled urbanization, and human migration. There are four serotypes of antigenically related dengue *flavivirus* (DEN 1 to DEN4), but immunity to one does not confer enduring immunity to another. The introduction of a new serotype in a determinate area accounts for the occurrence of epidemics and the hemorrhagic fever dengue form, which is a more severe and potentially lethal form of the disease.[63]

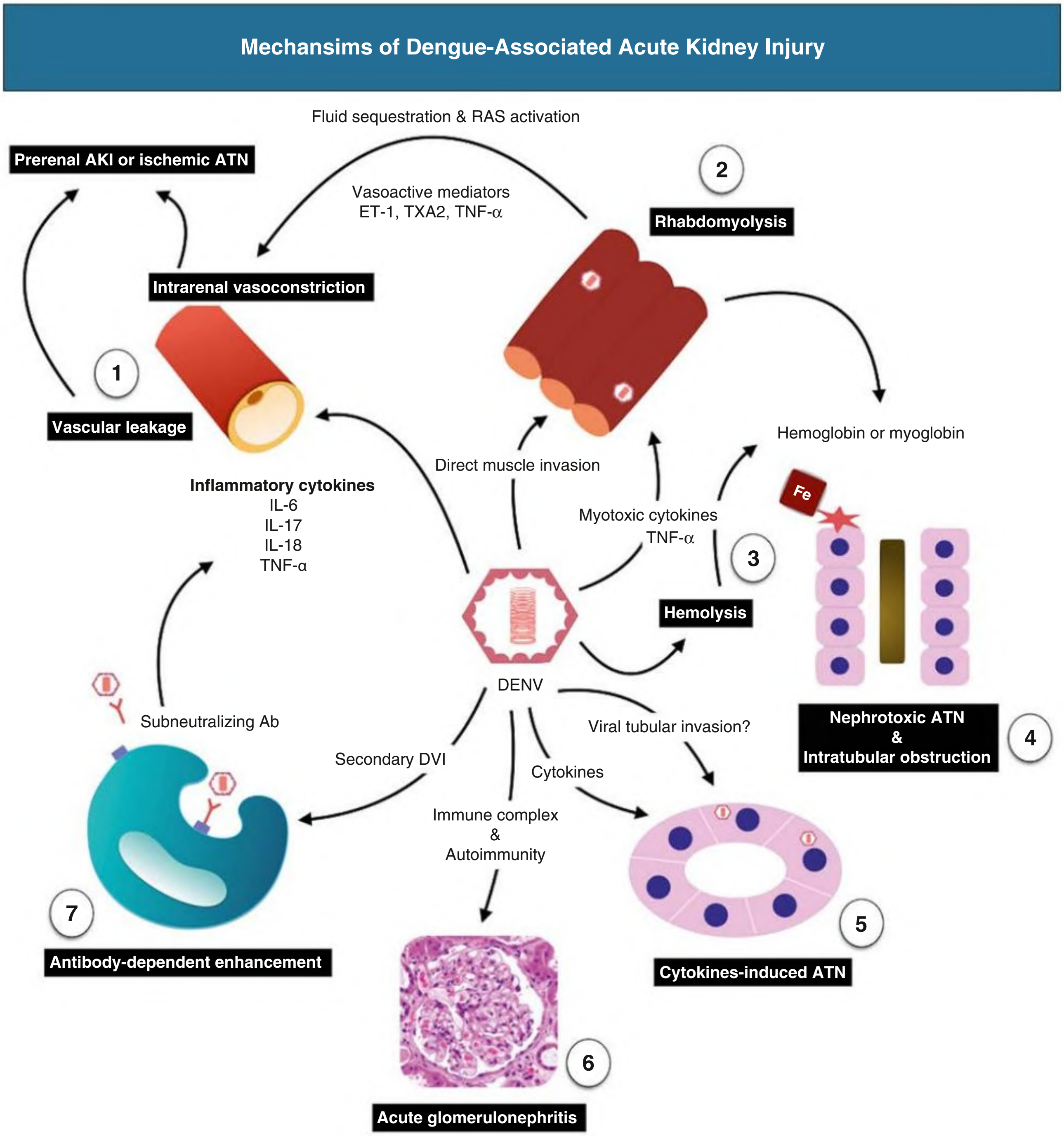

Fig. 71.11 Summary of Mechanisms of Dengue-Associated Acute Kidney Injury (AKI). (1) Dengue virus triggers cascades of inflammatory response, which lead to endothelial injury and vascular leakage. Reduced kidney perfusion occurs as a result. (2) Rhabdomyolysis in dengue virus is proposed to result from direct viral invasion and myotoxic inflammatory cytokines. (3) Intravascular hemolysis can be found, particularly in patients with glucose-6-phosphate dehydrogenase deficiency. (4) Myoglobin and hemoglobin are freely filtered into urine and directly injure kidney tubular cells. In some instances, they can, together with tubular proteins, form casts in the tubular lumens. (5) Acute tubular damage can be accounted for by inflammatory cytokines or by direct viral invasion suggested by the presence of viral antigen in tubular cells of dengue patients with AKI. (6) Evidence of dengue-associated acute glomerulonephritis with glomerular immune complex deposits has been reported, as well as development of autoantibodies to glomerular basement membrane (GBM) resulting in anti-GBM disease. The latter process is thought to be mediated by molecular mimicry of viral antigens. (7) Key immunopathogenesis of severe dengue virus infection (DVI) is based on antibody-dependent enhancement, where subneutralizing antibodies acquired from past infection promote dengue virus (DENV) entry into cells and suppress the antiviral response. *Ab,* Antibody; *ATN,* acute tubular necrosis; *ET-1,* endothelin-1; *IL,* interleukin; *RAS,* renin-angiotensin system; *TNF-α,* tumor necrosis factor-α; *TXA2,* thromboxane A2.

Dengue infection may manifest as undifferentiated fever, dengue fever, dengue hemorrhagic fever (DHF), or dengue shock syndrome (DSS). Common clinical manifestations of dengue fever are high fever, myalgia, arthralgia, retroocular pain, headache, anorexia, nausea, vomiting, and a cutaneous rash. DHF and DSS are severe forms of the disease, characterized by fever, hemorrhage, thrombocytopenia, and evidence of plasma leakage (increased hematocrit, pleural effusion, ascites, and hypoalbuminemia), mental disorientation, shortness of breath, tachycardia, shock, and death.[62,64]

Kidney involvement in dengue includes AKI, proteinuria (sometimes nephrotic), GN (see Chapter 57), hemolytic uremic syndrome, and electrolyte abnormalities.[62,64,65] The frequency of dengue-associated AKI varies from 1% to 30%, and its development and severity are related to poor prognosis.[62,64,66,67]

Dengue-induced AKI is usually associated with shock, hemolysis, and/or rhabdomyolysis but may occur without any of these triggers and is described in all forms of dengue.[62,64] Risk factors for AKI comprise increased hepatic enzymes, hypoalbuminemia, reduced serum bicarbonate, other simultaneous viral or bacterial infection, multiple organ dysfunction, use of inotropic or nephrotoxic drugs, increased age, obesity, dengue severity, rhabdomyolysis, presence of diabetes mellitus, and delayed hospitalization.[62,64]

The pathogenesis of dengue-associated AKI has been attributed to hemodynamic instability, rhabdomyolysis, hemolysis, direct viral invasion, and immune response. Patients with severe dengue infection, mostly reinfected by another viral strain, may develop plasma leakage syndrome, which eventually results in hypotension, causing kidney hypoperfusion and AKI. Inflammatory cytokines produced by dengue-infected monocytes and mast cells probably are linked to this phenomenon.[68] Rhabdomyolysis may occur rarely in dengue, and viral muscle invasion and inflammatory cytokines myotoxicity have been considered as possible agents causing muscles injury.[69] Data from autopsies and from studies in rodents demonstrated the presence of dengue virus in kidney tissue, suggesting direct kidney cytopathic effects. Host immune response may be involved in the pathogenesis of dengue-associated AKI. The concept of secondary infection as the key pathogenic factor for developing DHF/DSS has been well established for decades and is explained by antibody-dependent enhancement. Previous infection by one serotype of dengue virus results in development of subneutralizing antibodies. These antibodies possess high viral attachment efficiency enhancing internalization of virus into cells. Reports of DHF and AKI associated with proteinuria, hematuria, and reduced serum C3 level without presence of hypotension, rhabdomyolysis, hemolysis, or use of nephrotoxic agents support the role of possible immunopathogenic mechanism causing glomerular and tubular damage either by immune complex–mediated mechanism or cytokine-mediated tubular injury. Autoimmunity induced by molecular mimicry is another possible mechanism for dengue-associated kidney disease (Fig. 71.11).[62,64]

There is no specific treatment for dengue fever. Therapy is mostly supportive, avoiding the use of aspirin and nonsteroidal antiinflammatory drugs.

Yellow Fever

Yellow fever is a noncontagious infectious disease that is endemic in tropical Africa, South America, and Panama. The yellow fever virus is part of the *Flavivirus* genus (Flaviviridae family).[61] It is transmitted to humans by blood-eating insect bites, especially by the *Aedes* and *Haemagogus* genera. There are two cycles, termed *sylvatic* and *urban.* The sylvatic cycle affects sporadic individuals who come in contact with the vectors when performing economic or recreational activities in infested forests. The urban cycle is characterized by virus transmission through *A. aegypti* to individuals living in urban areas. The urban cycle was eliminated in the Americas from the 1940s to 1950s, but its resurgence was recently documented in Bolivia. The movement of viremic individuals to cities with high vector populations can potentially generate explosive urban epidemics affecting thousands of nonvaccinated people.[61] There is no evidence of yellow fever in Asia, despite an extensive presence of vectors.[70] It is possible that the hyperendemicity of dengue in Southwest Asia has afforded protection because of cross-reactive antibodies.

Yellow fever infection might be asymptomatic, cause moderate febrile disease, or be severe, causing hemorrhagic fever, liver failure, AKI, and death. Most patients (85%) fully recover after 3 to 4 days and become permanently immunized against the disease. About 20% develop the severe form, with mortality of up to 50%.[61] After 3 to 6 days of incubation, the clinical picture starts abruptly with high fever, chills, anorexia, myalgia, headache, vomiting, and bradycardia. Hemorrhagic manifestations may occur. There is then a remission period with symptom improvement, and mild cases do not have any further manifestation. In the severe forms, fever recurs followed by vomiting, epigastric pain, and jaundice—the so-called intoxication phase. There are large increases in transaminases and bilirubin, and leukopenia and ST segment abnormalities may develop. Hemorrhagic events associated with hepatic damage and consumptive coagulopathy can occur, such as hematemesis, melena, petechiae, bruises, mucosal bleeding, and metrorrhagia in women. Microcirculatory thrombosis, disseminated intravascular coagulation, tissue anoxia, oliguria, and shock may follow.[61]

Yellow fever–associated AKI occurs usually after 5 days of disease in the severe forms and may evolve to anuria and acute tubular necrosis, with increased mortality. In Africa, AKI is observed earlier, without jaundice or liver abnormalities, with higher mortality. AKI mechanisms are poorly understood. Kidney ischemia, intravascular coagulation, shock, bilirubin-induced kidney tubule cell toxicity, virus direct effect on kidney tissue, and intense inflammatory cytokine discharge are likely mechanisms of AKI.[71-73] In 2017 and 2018, outbreaks of yellow fever in Brazil AKI and increased kidney echogenicity were identified as independent factors associated to higher patient mortality.[74-75]

SELF-ASSESSMENT QUESTIONS

1. Bites from what families of snakes may lead to acute kidney injury?
   A. Viperidae and Elapidae
   B. Only Elapidae
   C. Only Tropidophiidae
   D. Colubridae and Aniliidae
   E. Only Aniliidae
2. A 25-year-old farmer from northeast Thailand, without underlying diseases, presented with high fever and history of back pain for the last 3 days. His vital sign revealed heart rate 120 beats/min, blood pressure 80/40 mm Hg, body temperature 38.8°C, and respiratory rate 30 breaths/min. Initial laboratory examination showed leukocytosis with neutrophil predominate. Urinalysis showed white blood cells numbering 10 to 30/high power field (HPF) and red blood cells 5 to 10/HPF.

   On the second day after admission, he was still febrile and had worsening of kidney function. Blood pressure was 120/70 mm Hg. His serum creatinine increased from 1.0 mg/dL to 4.2 mg/dL. A leptospirosis rapid test was negative. Indirect immunofluorescence assay for scrub typhus was pending. Dengue NS1 Ag was negative. Malaria thick and thin films were negative. Hemoculture was negative.

What is the most likely diagnosis?
A. Melioidosis
B. Scrub typhus
C. Dengue infection
D. Leptospirosis
E. Malaria

3. What is the correct sequence (from higher to lower frequency) of *Plasmodium* species associated with AKI development?
A. First *vivax,* then *falciparum,* less common *malariae, ovale,* and *knowlesi*
B. First *falciparum,* then *vivax,* less common *malariae, ovale,* and *knowlesi*
C. First *falciparum* and *vivax* similarly, then *malariae, ovale,* and *knowlesi*
D. *Falciparum, vivax, malariae, ovale,* and *knowlesi* similarly
E. First *malariae, ovale,* and *knowlesi* similarly, then *falciparum* and *vivax*

4. What kind of viral hemorrhagic fevers are more commonly associated with development of acute kidney injury?
A. Crimean-Congo hemorrhagic fever and Ebola
B. Chikungunya fever and Rift Valley fever
C. Dengue, yellow fever, and Hantavirus hemorrhagic fever
D. Malaria and leptospirosis
E. Tick-borne encephalitis and Brazilian hemorrhagic fever

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